**Quiz 5 - DSC 10, Spring 2025**

Full Name:

PID:

Quiz Time: ◯ 3PM ◯ 3:30PM ◯ 4PM ◯ 4:30PM

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**Instructions:**

- This quiz consists of 2 questions. You have a total of 20 minutes to complete it.
- Please write **clearly** in the provided answer boxes; we will not grade work that appears elsewhere. Completely fill in bubbles and square boxes; if we cannot tell which option(s) you selected, you may lose points.

  ◯ A bubble means that you should only **select one choice**.

  ☐ A square box means you should **select all that apply**.
- If your answer is a string, make sure to put it in quotes. If your answer is a float, make sure to include a decimal point.
- You may use one handwritten sheet of notes. No calculators and no computers.

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By signing below, you are agreeing that you will behave honestly and fairly during and after this quiz.

Signature:

# Version A

Please do not open your quiz until instructed to do so.

## Question 1

In music theory, **polyphony** is a number that counts how many independent melodies are being played at the same time. Polyphony is an integer at any given moment, but the average polyphony for a song can be a non-integer.

Minchan collects some data on pieces composed by Johann Sebastian Bach and Wolfgang Amadeus Mozart. Minchan's data is in the DataFrame `music`, whose first few rows are shown at right. For each piece, we have the composer and the average polyphony.

| | composer | polyphony |
|---|---|---|
| **0** | Mozart | 3.37 |
| **1** | Bach | 2.68 |
| **2** | Bach | 2.59 |
| **3** | Mozart | 2.84 |
| **4** | Mozart | 1.42 |

Minchan wants to conduct a permutation test with the following hypotheses:

- **Null Hypothesis**: The average polyphony of pieces composed by Bach and the average polyphony of pieces composed by Mozart are drawn from **the same distribution**.
- **Alt Hypothesis**: The average polyphony of pieces composed by Bach and the average polyphony of pieces composed by Mozart are drawn from **different distributions**.

For this permutation test, we will use **the absolute difference of medians** as our test statistic.

**a)** Fill in the blanks below to complete the permtuation test described above. Assume `observed` is the observed test statistic, and has already been calculated correctly.

```
total = 0
for i in np.arange(5000):
    shuffled_composer = np.random.permutation(__(a)__)
    shuffled_df = music.assign(shuff=shuffled_composer)
    poly = shuffled_df.groupby(__(b)__).__(c)__.get("polyphony")
    stat = __(d)__
    if stat __(e)__ observed:
        __(f)__
p_value = total / 5000
```

(a): 

(b): (c): 

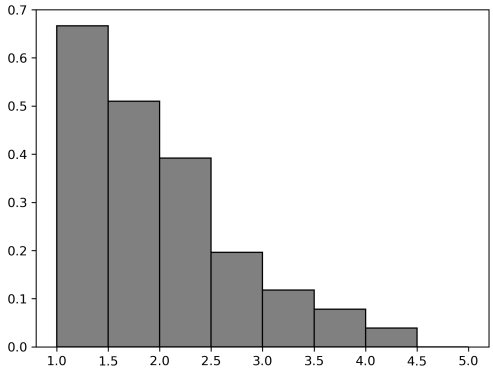
(d): 

(e): (f): 

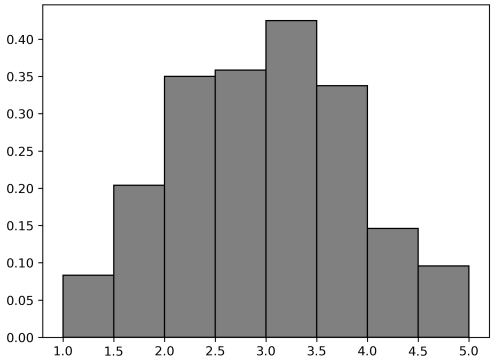
PID: _____

The following distributions come from the `"polyphony"` column of `music`:

**Bach's Average Polyphony**




**Mozart's Average Polyphony**




**b)** Based on these distributions, estimate the observed value of the test statistic in the permutation test above. Choose the closest answer below.

◯ $-3$ ◯ $-2$ ◯ $-1$ ◯ $0$ ◯ $1$ ◯ $2$ ◯ $3$

**c)** If Minchan repeats the permutation test, this time using **the absolute difference of means** as the test statistic, how will the observed value of the test statistic change?

◯ The observed value of the test statistic will be **smaller** than before.

◯ The observed value of the test statistic will be **the same as** before.

◯ The observed value of the test statistic will be **bigger** than before.

## Question 2

A Musical Instrument Digital Interface (MIDI) file is a digital file format that stores musical performance data. Jack has created a small MIDI file with 10 notes. For each note $i$ he has recorded two variables, $x_i$ and $y_i$.

- $x_i$ is 1 if the note falls on a down-beat, and 0 otherwise.
- $y_i$ is the MIDI velocity, an integer from 1 to 127 representing the force with which the note should be played.

Jack's data is below, along with some summary statistics.

| $i$ | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| $x_i$ | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 |
| $y_i$ | 78 | 86 | 86 | 86 | 78 | 86 | 78 | 78 | 78 | 86 |

$\text{Mean}(x) = 0.5$
$\text{SD}(x) = 0.5$
$\text{Mean}(y) = 82$
$\text{SD}(y) = 4$

**a)** Calculate $r$, the correlation coefficient between $x$ and $y$.

◯ $r = -0.6$ ◯ $r = -0.4$ ◯ $r = -0.2$ ◯ $r = 0.3$ ◯ $r = 0.5$

**b)** Calculate the slope of the regression line for the original dataset, using the value of $r$ you chose in part (a).

slope = ☐

**Reminders:**

- **Write your PID** on the front page and on the top right corner of page 3.
- If your answer is a string, make sure to put it in **quotes**.
- If your answer is a float, make sure to include a **decimal point**.
- Stay in your seat until the quiz is over. No leaving early.